Revising and Expanding Cushman's Learning-Based Model of Moral Cognition

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Abstract: Moral cognition refers to the human capacity to experience and respond to situations of moral significance. Recently, philosophers and cognitive scientists have turned to reinforcement learning, a branch of machine learning, to develop formal, mathematical models of normative cognition. One prominent approach, proposed by Cushman (2015), suggests that moral cognition is underwritten by a habitual ('model-free') system, in conjunction with socially-learned moral rules. I argue that moral cognition instead depends on three or more in decision-making systems, with interactions between the systems producing its characteristic sociological, psychological, and phenomenological features. Adopting such an approach allows us to not only better explain what is going on in everyday, 'successful' instances of moral judgment and action, but also to more reliably predict, and perhaps thereby counter, routine breakdowns in moral behavior.

Keywords: Moral psychology; moral cognition; formal models; reinforcement learning; domain-generality

1. Introduction

Moral psychology investigates how we experience and respond to situations of moral significance (Doris & Moral Psychology Research Group, 2010). Empirically-informed moral psychology proposes that the field be informed by relevant findings in the human sciences with an eye to advancing both descriptive and normative aims. The descriptive agenda involves

understanding and predicting everyday moral experiences and actions, often described as 'moral cognition.' The normative agenda includes intervening on and ameliorating the mechanisms underlying moral cognition with the goal of enhancing human happiness, well-being, and welfare. Ideally, cooperation on both of these goals should lead to bi-directional progress (Doris, Stich, Phillips, & Walmsley, 2017).

Empirically-informed moral psychology has made progress on elucidating the components of moral cognition (Greene, 2015). It is now widely accepted that affective experiences, and most notably emotions, have a profound influence on moral cognition (Nichols, 2004; Haidt, 2012; D'Arms & Jacobson, 2014). It has also been emphasized that we take factors such as consequences and personal harm into consideration when making moral judgments, supporting the conclusion that moral cognition involves rule-based inference and reasoning more broadly construed (May, 2018). Nonetheless, there is a growing sense that the field would benefit from a unified model of moral cognition.

Crockett (2016, p. 85) characterizes the current state of empirically-informed moral psychology in the following way:

> Imagine you want to bake a cake. The crucial first step is determining what ingredients are necessary for the cake—flour, sugar, eggs, milk, and so on. Next, you would need to know the amounts of each ingredient, and in what order to mix them. Similarly, past research in moral psychology has focused primarily on the critically important first step of identifying the key ingredients of moral judgments and decisions—norms, empathy, intentions, actions, outcomes, and so on. Now that many of these ingredients have been identified, future work can begin to develop formal mathematical models that describe

how they are combined, in what amounts and in what temporal order, to produce moral judgments and decisions.

Philosophers and cognitive scientists have turned to theories from computer science and cognitive neuroscience with the objective of developing such formal mathematical models of normative cognition. Perhaps most prominently, philosophers and cognitive scientists have turned to an area in computational neuroscience known as *reinforcement learning,* which studies how agents learn through interactions with their environments, to try to understand the nature of moral cognition (Shenhav and Greene, 2010; Crockett 2013; Cushman, 2013; Cushman, 2015; Crockett, 2016; Crockett, Siegel, Kurth-Nelson, Dayan, & Dolan, 2017).[1]

Reinforcement learning research suggests that human beings depend on at least three computational algorithms and, by extension, on three semi-autonomous decisions systems, for deliberation, choice and action. The *hardwired* ('Pavlovian') system relies on automatic approach, withdrawal responses to appetitive and aversive stimuli, respectively (Macintosh, 1983). The *habitual* ('model-free') system caches positive and negative state-action pairs. Finally, the *deliberative* ('model-based') system represents and selects from possible state-action pairs, often described in terms of a decision tree. Much of the present debate in this area centers on which of these systems best explain human moral judgments, choices, and behavior.

A prominent approach, proposed by Cushman (2015), suggests that moral cognition is underwritten by the habitual ('model-free') system, together with socially-learned moral rules. I argue, however, that although the model-free approach opens up a promising line of inquiry, it offers an unnecessarily narrow model of moral choice. That is, given that the reinforcement

[1] Reinforcement learning is an area in machine learning that aims to understand how agents learn to maximize rewards through interactions with their environments, i.e., how they learn through reinforcement, rather than through training on supervised data sets (Sutton and Barto, 2018).

learning literature indicates that all three decision-mechanisms continually trade off and interact to produce our everyday behaviors, there is no reason to accept Cushman's exclusive emphasis on the model-free decision system in the moral domain (Dayan & Niv, 2008; Dayan & Daw, 2008; Dayan & Berridge, 2014; Huys, et al., 2015; Garbusow, et al., 2018).

Building on Cushman's model-free account, I defend a new, multi-system model of moral cognition. Specifically, I argue that moral cognition depends on three or more decision-making systems, with interactions between the systems producing its characteristic sociological, psychological, and phenomenological features. Adopting such an approach allows us to not only better explain what is going on in everyday, 'successful' instances of moral judgment and action, but also to more reliably predict, and perhaps thereby counter, routine breakdowns in moral behavior.

I begin by sketching a set of features of moral cognition in need of explanation. I then go on to introduce the basic principles of reinforcement learning, before turning, in the main body of the paper, to the application of these principles to competing explanations of moral cognition. I conclude by considering some of the implications of my view for the normative agenda of intervening on human moral decision-making and, by extension, potentially ameliorating human welfare.

2. Moral cognition: What needs explaining?

Imagine a shopper named Barbara. Barbara is standing in the pasta aisle of her local grocery store, preparing to buy her usual brand of pasta. She then remembers that the company refused to feature a homosexual couple in its advertising.[2] Upon consideration, Barbara feels that

[2] This example is loosely based on controversial remarks made by Guido Barilla, of Barilla pasta, in 2013. The company has since improved its record on LGBTQ rights (Somashekhar, 2014).

it would 'just be wrong' to buy this brand and, seeing that a typically more expensive brand of pasta is on sale, she decides to buy this second type instead. Notably, Barbara remains unsure about which brand of pasta she will buy over the long term.

Barbara's pasta-buying experience is an example of *moral cognition*, or the capacity to experience and respond to situations of moral significance.[3] The experience raises a number of questions for empirically-informed theories of moral psychology. How do human beings make choices involving moral dimensions? Why do moral principles affect individuals differently at different times? And where does the feeling that so often accompanies moral choices, namely that something is just 'right' or 'wrong,' come from?

However, one challenge facing the empirically-informed, moral psychological study of moral cognition is that it is a complex and multi-faceted capacity. No one feature seems to get at the essence of the human capacity for moral experience. Moreover, moral cognition is not a regular mechanism in the way that other targets of explanation are reliable: while we may occasionally feel compelled to do what is right, we just as frequently abandon our moral commitments to do what is pleasant or even expedient (and as we will see in this section, this punctuated compliance has both psychological and phenomenological components). How best to proceed, then, in our investigations of such a challenging subject of inquiry? And, perhaps even more difficult, how to choose between competing theoretical explanations of it?

In this section, I characterize moral cognition using a cluster approach. That is, I propose to draw on extant, broadly agreed-upon analyses of moral norms, judgments, and emotions to

[3] By cognition, I mean the general functioning of the mind, rather than the narrower mental processes associated with understanding and knowledge. Moral cognition here refers to our capacity for lay moral experience, including the experience of moral emotions such as anger or shame. In other branches of philosophy, moral cognition is sometimes referred as *moral perception*; however, this latter term has technical connotations that I do not want to import in the discussion here (for a review, see McGrath, 2018).

characterize moral cognition in terms of its core sociological, psychological, and phenomenological features. I further propose to use these features to assess and adjudicate between competing theoretical alternatives. Here, the working assumption is that, the greater the number of features a given theory can account for, the better its standing vis a vis competing alternatives.

2.1. Sociological features of moral cognition

Philosophers typically emphasize the role of moral norms in our human capacity for moral cognition. But from more empirically-informed perspectives, moral norms, and norms in general, are in fact most commonly studied from a sociological perspective, particularly using game-theoretic approaches. On these latter methodologies, moral norms are sometimes said to represent a special class of the 'social grammar' of social norms: moral norms are the products of human interaction, and they dictate what is and is not morally appropriate for members of a given group or society (Bicchieri, 2006).

In the context of this sociological approach, moral norms are broadly considered to be ubiquitous and pervasive. There is evidence of moral norms in all past and present human societies. Further, Sripada and Stich (2005) suggest that norms follow reliable patterns of distribution, which is to say that norms cluster around general themes, such as prohibitions of killing and incest. At the same time, there is variability in how these normative themes become implemented into specific rules or 'maxims.' For example, what does and does not constitute acceptable grounds for a violent altercation can vary a great deal between different groups and communities. Typically, there also some exceptions to the norms held by a given group or society.

We can thus call *variable ubiquity* a sociological feature of moral cognition: moral cognition is reliably present among all human communities, although the contents involved in those moral cognition processes can vary to a relatively substantial degree.

2.2. Psychological features of moral cognition

There is growing moral psychological interest in our capacity to assess, cognize, and respond to moral circumstances, complementing the substantial sociological and game theoretic literatures on moral cognition. In addition to the sociological feature of variable ubiquity, this body of literature highlights four psychological features of moral cognition. First, the developmental literature indicates that moral cognition exhibits a *reliable pattern of ontogenesis*, which is to say that typically developing human beings reliably develop the capacity to process and engage with the moral components of experience. For example, children as early as three years of age appear able to distinguish between non-moral and moral concerns (Smetana & Braeges, 1990), to assess the role of intentions in assigning moral praise and blame (e.g., Nobes, Panagiotaki, & Pawson, 2009), and to protest when a third party is subject to moral transgressions (Rossano, Rakoczy, & Tomasello, 2011).

Second, moral cognition is widely thought to be *cognitively heterogenous*: it can be executed both quickly and intuitively, as when people automatically respond to an everyday moral circumstance, e.g., helping a person in immediate need. Alternately, moral cognition can be based on careful internal deliberation, as when people turn over their moral alternatives in their minds, particularly in facing a difficult moral choice.[4] This cognitive heterogeneity is often

[4] This latter, deliberation-based form of moral cognition is different from a still third kind of moral capacity, namely, moral reasoning or debate – the sort of reasoning on display in the pages of philosophy journals, where we aim to develop general and intermediate moral principles, defend and apply them, and so on.

characterized in terms of dual processing accounts (Haidt, 2012; Cushman, Young, & Greene, 2010; Campbell & Kumar 2012; Greene, 2013; May 2018), where moral cognition depends on both a fast, automatic, and emotional system, as well as on a slow, demanding, and reasons-based system. One prominent, dual-process account further linked the automatic, emotional system with deontological, or 'duty-based' moral responses, and the slow, reasons-based system with utilitarian, or 'consequences-based' moral reasoning (Greene, 2001; Greene 2014). For our purposes, in an effort to avoid over-theorizing the phenomenon in advance, we can describe this psychological feature of moral cognition in much more open-ended terms, namely, by simply recognizing that the capacity for moral engagement appears to rely on more than one form of cognitive processing.

Third, moral psychologists frequently argue that moral cognition is reliably associated with certain types of *emotions*. For example, Prinz and Nichols (2010) note that moral cognition is regularly associated with self-blaming emotions such as guilt and shame, and other-blaming emotions such as anger. Along similar lines, Kelly (2011) argues that the emotion of disgust plays a central role in moral cognition. Notably, one does not have to be a *sentimentalist* about moral cognition, that is, one does not have to hold that emotions are *constitutive* of our moral capacities, in order to recognize that there is a reliable association between moral responses and certain types of emotions (May 2018).

Fourth, moral cognition can be characterized by the aforementioned *irregularity* that we so often associate with moral behavior: that is, we typically comply with our moral principles and values, but there are not infrequent exceptions to this rule (Doris, 2002). In other words, as individuals, we take moral principles and rules to be compelling, although even a brief consideration reveals that, apart from social and punitive repercussions, nothing really obliged us

to do the right thing and that, indeed, we sometimes fail to do what is good. This last psychological feature of moral cognition represents a particular challenge for mechanistic explanations since, in principle, mechanistic explanations aim to account for patterns and *regularities* in a given phenomenon.

2.3. Phenomenological features of moral cognition

Finally, and in some respects most intriguingly, it may be said that moral cognition is associated with a distinctive phenomenological 'feel.' To be moved by what is right – or, conversely, to be moved to avoid doing what is wrong – is typically accompanied by a certain feeling of motivation or force or, as Cushman calls it, an accompanying feeling of '*moral constraint*' (2015). There is a special sense in which moral considerations move us in ways that other types of considerations do not. The feeling of moral constraint is thus a core feature of moral cognition that must be accounted for, and either be assimilated with or distinguished from other types of motivation.

The next section provides a brief overview of these reinforcement learning approaches, before turning to its application to moral cognition.

3. Reinforcement learning

3.1. Basic features

The reinforcement learning research program broadly refers to normative, computational approaches to understanding how agents learn to maximize rewards through interactions with their environments. Early theories proposed that learning only occurs when a single event violates an agent's expectations (Rescorla & Wagner, 1972; see also Schultz, Dayan, &

Montague, 1997). A subsequent theory (Sutton and Barto, 2018) developed this foregoing principle to analyze how agents learn to maximize their rewards over time. Here, we focus on Sutton and Barto's reinforcement learning framework, together with subsequent developments, to show how principles of the theory are used to analyze choice and action.

Reinforcement learning models generally consist of four main components: a set of states, a set of actions, rules governing the transitions between states, and rules that determine the immediate reward associated with a given transition and corresponding state. Notably, in reinforcement learning environments, an agent does not know how the transitions work, or what the rewards will be in each state. Instead, she must learn from experience and try to predict the values of transitions and rewards that she has not yet made. To do so, an agent can deploy one or more computational strategies to try and best estimate the value of different courses of action, and ultimately to arrive at a policy that will enable her to collect as much reward as possible.

The current consensus in the literature suggests that decision-makers approaching optimality – including human beings – rely on not one but at least three dissociable decision-making systems to make choices in their environment (for two excellent introductions, see Montague, 2006; Redish, 2013; for more specialized reviews, see Dayan & Abbot, 2001; Dayan & Niv, 2008; Rangel, Camerer, & Montague, 2008).

3.2. The three systems

We can now discuss each of the three systems in detail. Behaviors issued by the first of the three systems, known as the *Pavlovian* system, are characterized by automatic approach and withdrawal responses to appetitive and aversive stimuli, respectively (Macintosh, 1983). For example, we 'naturally' *approach* pieces of tasty food, but *avoid* scary snakes or spiders. These

general types of behaviors are thought to be underwritten by the evolutionarily old, 'hardwired' Pavlovian systems. Notably, though Pavlovian responses are appropriate in natural environments, since it is broadly beneficial to approach rewards and avoid punishments, these responses can also lack flexibility, which can result in detrimental outcomes (Huys, et al., 2012).

Due to this characteristic behavioral rigidity, researchers identify Pavlovian responses by persistence even in those cases where this ongoing, 'stubborn' response is detrimental. For example, David and Harriet Williams demonstrated that pigeons continue to peck at a key associated with food, even when doing so resulted in losing the reward (1969; see also Brown & Jenkins, 1968). Similar response patterns have been found with other animals, including in human beings (Sheffield, 1965; Herschberger, 1986; Bouton, 2006; Redish, 2013).

A second, *model-based* system explicitly represents possible choices and determines the sequence of actions that maximizes value (Dayan, 2011; Daw & O'Doherty, 2013). This procedure is typically represented by a *decision tree*. Each node in the tree represents a possible choice. The model-based system searches through the decision tree to find the branch with the highest total value. Hence its alternative name: tree search. For example, a chess player may represent three upcoming moves in a game of chess, with each possible move further branching into a wide range of subsequent moves. To win, the player then tries to represent and choose the best possible sequence of moves overall. Of course, representing all of one's future moves in chess is famously challenging, because the options quickly branch beyond any reasonable ability to explicitly represent them. Consequently, whereas Pavlovian responses are automatic and inflexible, the model-based system responds flexibly to new situations but struggles if there are more than a few options to consider.

A third, *model-free* system does not explicitly represent future alternatives, but caches positive and negative experiences, and assigns values to actions based on their previous outcomes (Dayan, 2011; Daw & O'Doherty, 2013). To cache experiences, the model-free system employs a feedback signal, which revises the system's estimates about the environment. Specifically, the model-free system is thought to be described by the so-called temporal difference (TD) learning algorithm, which enables an agent to weigh expectations against actual rewards and use the comparisons to optimize actions over time. If the reward is larger than expected, the signal indicates that things have gone 'better than expected'; if the reward turns out to be smaller than anticipated, the signal registers 'worse than expected'; if the anticipated and experienced rewards are the same, the signal records 'no change.' On this approach, 'good' state-action pairs are those that have produced rewarding outcomes in the past, and so should be repeated. Conversely, 'bad' state-action pairs have produced punishments in the past, and so should be avoided.[5]

In virtue of its caching-based approach, model-free learning exhibits the opposite computational strengths and weaknesses of those of model-based learning. As its name suggests, the model-free system does not apply the modeling stages of model-based learning. Rather, it interacts with the environment and updates its estimation of the best course of action over time. One important byproduct of this approach is that the model-free system is not immediately sensitive to unexpected change. Due to the 'caching' nature of the algorithm, it instead takes several low-reward experiences to lower the overall high values that had previously been attributed to a state. This 'recalcitrant' response is taken as a signature feature of model-free learning – a feature that will become important in our discussions of moral cognition. On the

[5] I owe this formulation to Crockett (2013).

other hand, over time, the system is efficient, because, unlike the model-based system, it does not need to explicitly represent all of the action possibilities in every given decision.

3.3. Linking the normative models to the brain

The greatest discovery linking normative reinforcement models to actual biological processes was Wolfram Schultz, Peter Dayan, and Read Montague's discovery that the TD learning signal accurately predicts the firing of dopamine neurons in relationship to unexpected rewards (Schultz *et al.*, 1997; though see Langdon, Sharpe, Schoenbaum, & Niv, 2017; Gardner, Schoenbaum, & Gershman, 2018). Montague describes the discovery as follows. Contrary to the then-accepted view that dopamine reflects experiences of pure, immediate reward, "Schultz noticed that dopamine neurons changes their activity when 'important' events happened, like a juice squirt, or the appearance of food, or even a sound in the laboratory that predicted that food or drink was about to be delivered" (2006, p. 108). This led him to the possibility that dopamine could correspond to a kind of prediction error. When he looked at Schultz's data, Dayan immediately "recognized a striking resemblance between dopamine neuron activity and error signals used in abstract reinforcement learning algorithms… it was an amazing match. The model showed that Schultz had discovered one of the central critic systems in the mammalian brain, and one that encoded its criticism in the delivery of dopamine" (Montague, 2006, p. 109). Over the subsequent decade, findings have shown that all three systems have been found to play a prominent predictive role in decision-making and action, including in human decision-making and action (for helpful overviews, see Dayan, 2014; Dayan & Berridge, 2014; Daw & Tobler, 2014; Moutoussis, et al., 2018).

One primary, experimental way of distinguishing between the model-free and model-

based decision systems is known as *post-training reward devaluation* (henceforth 'reward devaluation') (Dickinson & Balleine, 2002; see also Tricomi, Balleine, & O'Doherty, 2009). In reward devaluation, an agent is trained to perform a certain action in exchange for a reward, e.g., pushing a button in exchange for a treat. The period of this training is either short, such that the agent continues to rely on her forward-looking, model-based decision system; or it is somewhat longer, such that the agent comes to rely on her caching-based, model-free system, i.e., that she can habituate the relevant activity. In experiments using this paradigm, the reward is then devalued, e.g., in the case of rats, by pairing it with a nausea-inducing chemical or, in the case of human participants, simply by overfeeding it to them, to the point where the reward is no longer pleasant to consume, and the agent's willingness to perform the task is then analyzed. In those cases where the training period was sufficiently *short*, the agent can respond 'appropriately' in the sense of no longer performing the action that generates more of the now-unpleasant reward. But in those cases where the training period was *longer*, and thus led to dependence on the model-free system to perform the instrumental action, the agent responds 'inappropriately,' that is, she continues to perform the task, even though the reward is no longer appealing her.

Emerging lesion studies further complement these findings. For instance, the orbitofrontal cortex has been shown to a particular role in representing model-based values (Valentin, Dickinson, & O'Doherty, 2007). Lesions to the dorsolateral striatum and nigrostriatal dopamine systems prevent an animal from developing model-free learning (Yin, Knowlton, & Balleine, 2004; Faure, Haberland, Conde, & El Massioui, 2005). And while relatively little is known about the structures underlying potential arbitration between the two valuation systems, some evidence is beginning to emerge. In particular, two candidates for arbitration are being proposed in the shape of the infralimbic cortex and the anterior cingulate cortex. Lesions to the

infralimbic cortex reintroduce model-based learning from what had previously relied on the model-free controllers, and studies have shown that anterior cingulate cortex is involved in in monitoring and resolving response errors and conflict (Daw, Niv, & Dayan, 2005, p. 1708).

Admittedly, having multiple systems competing to evaluate a single task might seem at best inefficient and at worst disadvantageous. However, simulations suggest that optimal agents in fact employ several complementary controllers (Dayan 2011; Daw, et al., 2005). As Daw and colleagues note, "The difference in the accuracy profiles of different reinforcement learning methods both justifies the plurality of control and underpins arbitration. To make the best decisions, the brain should rely on a [system] of each class in circumstances in which predictions tend to be most accurate" (2005, p. 1704).

One line of inquiry has analyzed how a possible *principle of arbitration* can dictate the circumstances under which a given system should govern a specific choice problem (Daw, et al., 2005, Lee Shimojo, & O'Doherty, 2014). A prominent approach proposes that interactions between the systems are governed by the following 'accuracy-based principle of arbitration' (henceforth, 'PA') (adapted from Daw, Niv, & Dayan, 2005):

PA: Following partial evaluation, that system with the highest accuracy profile, i.e., that system most likely to provide an accurate prediction of expected value, relative to the decision problem at hand, directs the corresponding assessment of value.

On this view, each of the multiple systems partially evaluates the action alternatives. Simultaneously, each system generates an estimate of how accurate its prediction is relative to the decision problem at hand. These estimates, or accuracy profiles, are then compared, and the system with the highest accuracy profile is selected to direct the corresponding valuation task.

PA thus directs valuation based on a system's accuracy profile, rather than on its prediction of value (Deneve & Pouget, 2004; Daw, et al., 2005; Lee, et al., 2014). For example, the Pavlovian system typically coordinates choice in familiar, complex settings, because it typically has a higher accuracy profile in those decision problems, even if it predicts a lower overall value than do either its model-based or model-free counterparts.

A related line of inquiry suggests that all three decision-mechanisms continually trade off and interact to produce even a single, everyday behavior (Dayan and Niv 2008, Dayan and Daw 2008). For example, it is thought that the model-based system may be used to train up its model-free counterpart, while some evidence suggests that both the Pavlovian system is systematically used to 'prune' branches in model-based decision-trees in order to overcome the latter's signature computational complexity (Huys, et al., 2012).

Taking both of the lines of inquiry together, a picture emerges on which all three systems continually interact and trade off to produce our everyday choices and actions. In what follows, we will see that this rich network of interactions may help explain the multi-faceted and sometimes perplexing nature of everyday moral cognition. But first, let's look at Cushman's influential (2015) proposal regarding reinforcement learning, model-free learning, and moral cognition.

4. Cushman's model-free approach to moral cognition

4.1. The core thesis of Cushman's view

Fiery Cushman (2015) considers the mechanisms underpinning human moral cognition with a special emphasis on moral constraint. Cushman suggests that the question of moral

constraint is of particular interest insofar as the feelings associated with moral cognition seem to compel us in ways that other kinds of considerations do not. Cushman argues that these feelings and the corresponding behaviors are underwritten by the model-free decision system.

Cushman makes his case by drawing on the signature feature of model-free learning. Recall from the previous section that model-free learning does not represent future alternatives, but instead caches positive and negative experiences and predicts future value accordingly. Recall also that this caching procedure means that the system is not immediately sensitive to unexpected changes. The central premise in Cushman's argument is that moral behaviors exhibit precisely this signature feature of model-free learning, namely, this lack of sensitivity to changes in reward value. Hence, moral actions, including altruistic actions, are broadly thought to be supported by automatic, i.e., model-free responses (Rand, et al., 2014). Specifically, Cushman argues that when the brain produces the affective component of moral cognition and constraint, such as a feeling of obligation, "we assign intrinsic value to the action, rule or norm in question," such that it is difficult to act otherwise than in accordance with it (2015, p. 60-61). For example, Cushman argues, American tourists frequently continue to tip in restaurants abroad, even when there is no relevant norm dictating that they should do so (2015, p. 59). Related examples include apologizing where an apology is not really necessary, or quickly helping a person in need without first stopping to think about what the right thing to do would be.

Cushman illustrates his thesis by drawing on the model-free explanation to explain participants' inconsistent responses to the trolley problem. Famously, in 'switch' versions of the trolley problem, where the cause of a hypothetical harm is relatively indirect, participants are more likely to support the killing of a single individual in order to save five others. Inconsistently, however, participants are more likely to find it difficult to endorse the harm of

one agent in 'footbridge' versions of the problem, where the harm involves pushing another individual and so is more 'hands on' (Greene, 2001).

Since a purely numerical assessment favors the saving of five people rather than one in *both* cases, Cushman reasons, people's tendency to resist harming the single agent in the footbridge version may be "the consequence of," and so explained by, "negative value assigned intrinsically to an action: direct, physical harm" (2015, p. 59). That is, Cushman suggests, participants' responses to the footbridge version of the dilemma may be underwritten by the model-free decision-system. Since directly harming others has reliably elicited punishments in the past, this option represents a bad state-action pair, on the model-free view, and leads people to reject it as an appropriate course of action.

4.2. Assessing Cushman's view with respect to the features of moral cognition

How does Cushman's model-free view fare with respect to the sociological, psychological, and phenomenological features of moral cognition outlined above?
On my view, Cushman's account has both significant explanatory advantages and important explanatory disadvantages. The proposal's central advantages stem from its appeal to the domain-general, model-free decision system. A domain-general decision system is one which can be applied to all kinds of decisions, e.g., ranging from economic choices between goods to choices about how best to act. Explaining moral cognition in terms of such a domain-general system is advantageous because it avoids positing a dedicated system or mechanism for explaining our moral experiences. As Greene (2015, p. 40) observes, Cushman's account joins other accounts in suggesting that "what we call 'moral cognition,' is just the brain's general-purpose cognitive machinery— machinery designed to learn from experience, represent value

and motivate its pursuit. represent mental states and traits, imagine distal events, reason, and resist impulses – applied to problems that we, for high- level functional reasons, identify as 'moral.'?"

Cushman's domain-general approach also helps explain moral cognition's characteristic ubiquity, ontogenesis, and motivational phenomenology. We can start with the ubiquity of moral cognition. As noted in Section 2, moral cognition is thought to be a ubiquitous feature of human experience. But ubiquity is precisely what we should expect if moral cognition depends on a much older, more general cognition system such as the decision-making system. Moreover, if this approach to moral cognitive ubiquity is right, then we should also expect that *breakdowns* in moral cognition are associated with related breakdowns in the more general capacity, namely, in this case, in the more general capacity to make decisions.

This is indeed what we find. Empirically-informed moral psychologists commonly cite psychopaths as paradigmatic examples of individuals who fail to exhibit typical moral cognitive capacities, because they seem in many in ways incapable of making typical moral judgments and decisions (Blair, 1995; Nichols, 2002; Blair, 2017; though see Javala & Griffiths, 2017). But psychopaths exhibit corresponding, general deficits in decision-making, for example, in their performance on decision-making tasks such as the Iowa Gambling Task (Mahmut, Homewood, & Stevenson, 2008). In this regard, Cushman's proposal that moral cognition is supported by the model-free decision system falls squarely within a growing body of literature regarding the relationship between moral and general decision-making (Shenhav & Greene, 2010). Of course, this does not yet suggest that moral cognition *only* involves domain-general decision systems – only that these systems at least play a role.

An analogous line of reasoning can be presented regarding the reliable ontogenesis of moral cognition. If moral cognition is underwritten by the model-free system, and so depends on this more general decision-making capacity, then we should expect children's moral capacities to 'come online' at around the same time that they begin to exhibit adaptive, automatic decision-making abilities, e.g., on modified versions of the Iowa Gambling task. And indeed, although the data is less clear here than in the case of psychopaths, findings do suggest that children do begin to make advantageous choices on decision tasks at around the same time that they begin to recognize the moral features of their environment (Crone & van der Molen, 2004; Killen & Smetana, 2015).

Similarly, finally, the domain-general nature of the model-free decision system explains the characteristic phenomenological experience of moral constraint. It is generally thought that reinforcement learning, including the model-free system, explains how choices are transformed into corresponding actions, i.e., how choices motivate action (Glimcher & Fehr 2013). Correspondingly, if we think of moral cognition as supported by this more general system, then we can equally expect it to explain the phenomenology associated with specifically *moral* motivation. Indeed, one of the central implications of Cushman's view is that moral motivation may operate in roughly the same way it does in other decisions, albeit perhaps more forcefully (Cushman, 2015, p. 60).

On the other hand, the view's exclusive emphasis on the model-free decision system limits its overall explanatory power. That is, the model-free view struggles to explain moral cognition's cognitive heterogeneity, punctuated compliance with moral norms, and reliable association with moral emotions. Given that Cushman's view exclusively focuses on model-*free* learning, it struggles to explain how we might explicitly represent or deliberate about a given

moral dilemma. Similarly, if we think of the model-free system as supporting something akin to habituation, it is hard to reconcile this computational approach with the observation that we routinely – but irregularly – fail to act in ways that correspond with our moral norms and judgments. And the model-free view is simply *neutral* when it comes to presence of moral emotions in moral cognition. That is, the model-free view is neither at odds with this feature of moral cognition, nor does it have any specific resources to explain why these emotions are so reliably present.

4.3. *Two objections and a proposal*

Despite these explanatory limitations, the model-free view of moral cognition represents a highly promising alternative to extant, highly influential dual-system views of moral judgments and decision-making (Driver, 2016). Nonetheless, Cushman's account faces two more general objections.

First, it is important to recognize that all three of the view's foregoing explanatory successes rely on the domain-general nature of model-free learning, rather than on the specific features of the model-free system itself. This is not problematic in and of itself; but it is worth recognizing that either or both of the other two decision systems share this explanatory power. That is, the model-free explanation of ubiquity, ontogenesis and moral constraint is more generally a *reinforcement learning-based* explanation, and so is not exclusively available, in explanatory terms, to the narrower view Cushman puts forward.

Second, I argue that Cushman's proposal is simply unnecessarily restricted in its use of the reinforcement learning toolkit. That is, given the growing body of evidence which suggests that all three decision-mechanisms continually trade off and interact to produce our everyday

behaviors, Cushman's exclusive emphasis on model-free learning is problematic (Dayan & Niv, 2008; Dayan & Daw, 2008; Dayan & Berridge, 2014; Huys, et al., 2015). Correspondingly, it seems implausible that normative actions alone are exclusively underwritten by the model-free mechanism, as Cushman's view suggests.

We can illustrate this latter point by applying it to Cushman's assessment of the trolley problem. In particular, more evidence would be needed to establish that it is exclusively the model-free system that is involved in participants' responses to the footbridge version of the trolley problem. This is because, if we recall from above, *two* systems are known to respond sub-optimally in the face of a change in rewards: both the model-free *and* the Pavlovian systems respond in this way (see Crockett, 2013). The difference is that while the Pavlovian system is *permanently* insensitive to reward devaluation, the model-free system slowly adapts to reward devaluation over time (albeit much more gradually than its model-based counterpart). Within the reinforcement learning framework, there are thus *two* plausible interpretations of peoples' responses to the physical harm involved in the footbridge dilemma. On Cushman's hypothesis, the participants' aversion to harm is rooted in the model-free valuation of a habitually negative state-action pair. Alternatively, the participants' aversion may be underwritten by a much more deeply rooted, Pavlovian response to overwhelmingly aversive stimuli (Huys, et al., 2012).

One way to disentangle these competing theories would be to conduct a targeted version of the trolley experiment, designed to test for the role of either the model-free or Pavlovian learning systems in people's responses to the problem. Using a head-mounted display device, the participants would be placed in a virtual reality environment (VRE) that imitates classic trolley problem scenarios (VRE-based methodology adapted from Navarrette, McDonald, Mott, &

Asher, 2012). Participants would be told they have two alternatives to choose from in each simulation: to stop an oncoming train, or to do nothing.

The participants would be divided into two conditions. In the *switch condition,* a participant would appear to be standing near a railway track, with five railway employees shown to be repairing the tracks further on down the main line. There would also be a switch next to the participant's right hand (physically represented by a joystick in the lab). Left as is, the switch would allow a virtual train to continue straight on the main track, killing the five employees. If pulled, the switch would divert the train onto a sidetrack and kill a single individual. In the *footbridge condition*, the participant would appear to be standing on a footbridge spanning a railway track, while the five railway employees will be shown repairing the tracks below. In addition, a virtual pedestrian would be standing on the footbridge directly in front of the participant. Pushing the fellow pedestrian onto the track, physically represented by a soft mannequin in the lab, would cause the train to stop before hitting the five employees. Not pushing the pedestrian onto the track would result in the train continuing down the main track and killing the five individuals. In both conditions, sounds of distress coming from either the one or five agents would become audible, depending on the participant's choice.

The key feature of the experiment would consist in repetition. Recall that the Pavlovian and model-free systems are both 'slow to update.' But this is not quite accurate. The model-free system is slow to update; the hardwired, Pavlovian system *never updates at all*. Hence, the participants would be asked to experience multiple versions of both conditions. If Cushman's hypothesis is correct, the participants' responses to the footbridge condition would gradually update to reflect more 'utilitarian' responses. If the alternative hypothesis is correct, the

participants' responses to the footbridge condition would remain consistent in their refusal to push the imaginary individual off of the footbridge to save the five individuals on the tracks. In the next section of the paper, I propose a view that accommodates both hypotheses.

5. A multi-system view of moral cognition

At the outset of this paper, I proposed to use Cushman's model-free account as a jumping off point for a novel, multi-system, reinforcement-learning-based model of moral cognition. We are now in a position to take this step. In contrast to Cushman (2015), I argue that the Pavlovian, model-based, and model-free systems are all implicated in normative cognition.

5.1. The multi-system view

Many of the pieces of the multi-system view are already in place. It stipulates that moral cognition is underwritten by three domain-general, decision systems: namely, the Pavlovian, model-based, and model free decision systems. As we will see in what follows, many of the core features of moral cognition reflect the signature profiles of these three decision systems. In addition, these three systems interact to produce additional types of decision-making responses. Interactions between these systems are governed by the principle of arbitration characterized above, namely, PA. These interactions can take two general forms: first, PA may be used to a 'select' a 'winning' decision system, which goes on to make a corresponding, valuation-based decision; or, alternately, second, PA can dictate that two or more systems trade off on a single decision, as when the Pavlovian system prunes the model-based decision tree in order to simplify the choice matrix.

Importantly, by taking all three systems into consideration, together with the ongoing interactions between them, the multi-system model can explain the six core features of moral cognition.

5.2. Assessing the view

In addition to Cushman's emphasis on the model-free system, the forward-looking nature of model-based learning enables us to explain the deliberative components of moral cognition and, by extension, its cognitive heterogeneity. Moral reasoning often involves such capacities as setting goals, deliberating about what it would be best to do, and reconciling competing moral demands. These are the signature features of model-based learning: an agent identifies a specific goal, evaluates the different available means of achieving it, and pursues what she has identified as the best alternative. At the same time, the computationally taxing nature of model-based learning may help explain the use of explicit deliberation mainly in relatively novel or high-stakes moral situations, rather than in more familiar or low-stakes settings, as in Cushman's (2015) example of tipping in a restaurant, discussed above. The multi-system account thus provides a comprehensive approach to understanding the heterogeneous nature of moral cognition.

For their part, given their rapid, hardwired, and recalcitrant nature, Pavlovian responses most obviously figure in moral cognition as emotional responses. A wide range of theories highlights the importance of various emotions in normative behavior, emotions including guilt, shame, and anger (for a review, see Prinz & Nichols, 2010). But emotional responses are thought to themselves be a basic type of Pavlovian action, triggering automatic changes in heart rate, facial expression, and so on (Redish, 2013).

From a slightly different perspective, recall that Pavlovian responses are characterized by the fact that they persist well beyond optimality, as when chickens continue to peck at a dispenser even when it means not receiving their food. But this is precisely the kind of response we see in certain moral social interactions. For instance, experiments based on the ultimatum game, which ask participants to accept or reject monetary offers from a confederate researcher, show that participants frequently reject offers that they perceive to be unfair, even if doing so prevents them from receiving any money at all (van't Wout, Kahn, Sanfey, & Aleman, 2006). These decisions can thus be seen as a form of the recalcitrant or suboptimal decision characterized by Pavlovian responding.

More controversially, Pavlovian responding may also play a role in the phenomenon known as *moral dumbfounding*. Moral dumbfounding can be defined as the "stubborn and puzzled maintenance of a judgment without supporting reasons" (Haidt, Bjorklund, & Murphy, 2000, p. 1). For example, in a classic example of the phenomenon, subjects continue to believe that sex between siblings is wrong, even when they are told that all potential sources of harm have been accounted for. Insofar as these subjects maintain their judgments even after it no longer makes sense to do so, however, their responses exhibit the characteristic feature of Pavlovian decision-making.

Contra Cushman (2015), the properties of all three decision systems thus play a role in explaining the core features of moral cognition. However, I argue, finally, that it is the *interactions* between systems that explain the irregular or punctuated nature of compliance with moral judgments and beliefs. We often think of moral cognition as operating smoothly until 'something goes wrong.' That is, the multi-system view suggests that these intermittent lapses may instead be the products of the systems' regular workings (Haas, 2018). The multiple

decision systems operate in parallel, trading off to optimize decision-making. But in turn, these tradeoffs may occasionally result in responses that do not accord with our deliberative judgments of what it is best to do, or even with what, based on our habituated, model-free systems, we *typically* do. These tradeoffs can occur when the model-free system overrides its model-based counterpart or, alternately, when the Pavlovian system prunes the decision tree in order to simplify the decision process. As a result, we most often feel compelled by our moral principles and judgments, but also not infrequently fail to act in accordance with them.

6. Conclusion

Moral cognition is implicated in discussions of normative considerations such as moral dilemmas (Nichols & Mallon, 2006), moral responsibility (Nahmias, Shephard, & Reuter, 2014), moral luck (Cushman, 2008), and the nature of the moral self (Strohminger & Nichols, 2014). Nonetheless, it remains difficult to understand how best to effectively *intervene* on moral cognition and, by extension, on these explicitly normative issues.

In my view, the multi-system model offers a promising way of investigating possible interventions in moral cognition. This is because there is a close connection between understanding a mechanism and knowing how to intervene on it. As suggested by Crockett's cake analogy (2016, p. 85), computational approaches to moral cognition are making important headway in terms of providing such 'recipes' – or mechanistic accounts – of how variously theorized 'ingredients' or components of moral cognition interact to produce our everyday human moral experiences. Such computational approaches can make increasingly accurate predictions about patterns of *breakdowns* in moral cognition, such as the relationship between the neurotransmitter serotonin and judgments about harm aversion (Crockett et al., 2010). By

extension, computational approaches to moral cognition can offer important clues to those who would leverage these mechanisms with the goal of intervening on and improving human moral decision-making.

The multi-system model of moral decision-making extends this computational and mechanistic approach to moral cognition. In particular, the multi-system approach helps explain how our moral decision systems interact, and so enables us to recognize patterns of suboptimal decision-making that are produced as a consequence of these interactions. As I have argued elsewhere, in this sense, our decision systems and their interactions can help us understand the "fault lines" of our decision-making and, specifically, of our moral decision-making (Haas, 2018). By extension, the multi-system model thus provides the preliminary foundations for a comprehensive set of principles to guide our moral choices and actions – and potentially begins to empower us to counteract such suboptimal responses if and when we sit fit to do so.

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